

ACSM Expert Consensus Statement on Weight Loss in Weight-Category Sports

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Abstract

Weight-category sports are defined by the requirement of a weigh-in before competition to provide performance equity and reduced injury risks by eliminating size discrepancies. Athletes in these sports try to gain a theoretical advantage by competing in weight divisions that are lower than their day-to-day body mass (BM), using a combination of chronic strategies (body-fat losses) and acute manipulations over a period of hours to days before weigh-in ("making weight"). Strategies to support safer practices include minimal competition weight classification based on preseason body composition, reductions in the period between weigh-in and competition, and prohibition of unhealthy weight loss techniques. At an individual level, expert guidance by a sports nutrition professional can help an athlete to establish a pragmatic and long-term approach to BM management, recognizing the nuances of their sport, to achieve favorable outcomes for both health and performance.

The sports include combat (*e.g.*, boxing, martial and mixed martial arts [MMA], wrestling), weightlifting, powerlifting, sprint football, and rowing. An exception is in horse racing, where the weight restriction on the jockey provides a handicap to the performance of the horse. The prevailing culture in such sports is a desire to compete in a category that is lower than their natural or day-to-day training body mass (BM), to gain a real or perceived advantage over lighter opponent(s) (2–4). Concerns over health risks and the fairness of competition underpin the efforts of many weight-category sports to enact rule changes and education programs, in the hope of reducing the prevalence and

magnitude of BM manipulation activities (5,6). It may seem convenient to consider weight-category sports as a collective because of their common need to achieve a specified BM prior to competition. However, recent research has highlighted the specificity and nuances of BM manipulation practices within each sport and the need for practitioners to support pragmatic strategies for management of BM around competition to optimize performance while safeguarding health (2–4). This consensus statement provides a summary of factors that should be considered and replaces the 1996 ACSM Position stand on Weight Loss in Wrestlers (7).

Introduction

A number of sports involve competition in weight categories or divisions (hereafter called weight categories), with the main aim of creating an even playing field and safer competition by matching competitors with similar physical characteristics (1).

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Characteristics across Different Weight-Category Sports That Influence Weight-Making Practices

In weight-category sports, athletes have their BM verified at an official "weigh-in" prior to competition to ensure they meet the specified weight requirements. "Making" or "cutting" weight (hereafter called making weight) involves a range of acute and/or chronically applied strategies. The perceived or actual benefits of competing in a lower weight category vary between sports according to the characteristics that determine success (4,6,8). Although there are some common weight-making practices across all weight category sports, specific protocols are often determined by the rules, customs, and characteristics that influence opportunities to acutely reduce BM prior to the weigh-in,

and subsequently recover from this in the period between weigh-in and competition. Table 1 summarizes some of the features of weight-category sports, with brief comments about how these characteristics may influence weight-making practices. Since rule changes can occur in these sports, this information is provided principally to illustrate how various characteristics can influence weight-making per se, or its effect on health and performance.

Key issues influencing weight-making practices include the number of weight categories available for competition, and the frequency with which the athlete needs to make weight (*e.g.*, frequency of competition and the number of times the athlete must weigh in over the course of an event). While an increased number of weight categories may provide more opportunities for an athlete to find a class that is best suited to their natural physique, the smaller weight increments between divisions also may create greater temptation to “cut” to a lower division. Sports that provide greater challenges include those with few divisions (*e.g.*, lightweight rowing), and those in which the weight categories vary according to the competition. For example, in many combat sports, there are fewer or different weight categories on the Olympic Games program than on other national or international competitions (see Table 1). Sports involving weekly competition (*e.g.*, horse racing and collegiate wrestling during the competitive season) require frequent perturbations in BM, whereas in others (*e.g.*, professional boxing or lightweight rowing), the athlete may only compete “at weight” a couple of times a year. For further insight into characteristics that shape weight making practices of athletes in weight category sports, the reader is referred to the following references: (2–4,22).

The timeframe between weigh-in and competition varies between sports, affecting the ability of the athlete to recover from any weight-making strategies. Horse racing represents the greatest challenge since successful jockeys must reweigh after the race, minimizing their opportunities to consume food or fluid before competition (23). At the other end of the spectrum, some sports (*e.g.*, professional boxing and MMA) offer substantial recovery time (up to 32 h) between the weigh-in and event, with almost unlimited opportunities for preevent fluid and food intake. While such sports allow the athlete to be well-fueled and hydrated prior to competition, these conditions support a culture of extreme BM manipulation (24), as evidenced by mean BM gain of up to 10% and individual values of 20% BM gain during the recovery period (25–28). Indeed, there may be a link between the amount of postweigh-in rapid weight gain (RWG) and the duration of the recovery period (4,29). Some sports (*e.g.*, judo and taekwondo) have tried to circumvent this by introducing a second weigh-in on the morning of the event for randomly selected competitors, with rules prohibiting a BM gain >5% from initial weigh-in. Other sports (*e.g.*, lightweight rowing, Olympic boxing) routinely implement a pretournament weigh-in plus repeat weight checks in the hours prior to any subsequent event in which the athlete competes. Unfortunately, there are anecdotal observations that such changes merely encourage the athlete to limit their recovery, rather than curtail the magnitude of the weight-making (30).

Although cultural beliefs and traditions underpin the weight making practices observed in many sports (3,4,22,24,31), characteristics of performance also may influence the degree to which they persist, because of the advantages of having a greater “true”

BM. In sports like rowing, there are significant advantages to having greater muscle mass/power and longer limb “levers” (32). Indeed, despite the considerable physiological stress of a race, often overlaid with thermal stress, and even though the rules of competition try to limit weight-making by implementing serial weigh-ins close to the time of competition, it appears that successful lightweight rowers do not “naturally” fit below their weight targets (31). Furthermore, there is evidence that lightweight rowers can preserve performance via planned nutrition support around some degree of weight making (31,33,34). On the other hand, a recent study of jockeys in Great Britain revealed that changes in the historical handicaps for horses have not kept pace with generational increases in BM, meaning that current weight limits are too low for almost all jockeys (35). There may be subtle differences between combat sports regarding the performance advantages of having a larger natural BM, as demonstrated by a correlation between success and the magnitude of RWG (8). It should be noted, however, that although the magnitude of RWG is often used as a surrogate measure of the degree of rapid weight loss (RWL), it may not be an accurate indicator (4). Nevertheless, in grappling sports like wrestling and judo, a larger BM and muscle mass/power may confer performance advantage (36), whereas in striking sports, such as taekwondo and boxing, greater stature and limb length provide an advantage that is only partially correlated with BM (37,38). Meanwhile studies in combat sports involving striking and grappling (*e.g.*, MMA) have failed to show a correlation between RWG and competition success (39,40). Aside from any direct competitive advantage, the culture of weight-making is entrenched by perceived psychological benefits around sport identity and the feeling of focus, commitment, and self-confidence (41).

Common Weight-Making Practices

For the purposes of this position stand, chronic BM management includes strategies implemented over several weeks or months to facilitate a change in fat mass (FM) and fat-free mass (FFM) (sometimes termed interchangeably as lean body mass). Here, the athlete aims to reduce FM while maintaining or gaining FFM. Conversely, RWL (sometimes also called acute weight loss or AWL) includes the intentional manipulation of total body water, glycogen stores and gastrointestinal tract contents over a period of hours or days. Typically, RWL has limited impact on FM or FFM, although changes in body water may cause artifacts in assessments of body composition via various modalities and some loss of fat-free dry solid mass can occur in as little as 4 days (42,43). While specific BM management practices vary according to sport and athlete caliber, the majority of athletes engage in both chronic and acute strategies (3,22,24,25,29,31,44).

The 1996 ACSM Guidelines (7) advocated an emphasis on chronic BM management practices, including gradual dieting and manipulation of training load, while promoting a cautious approach to acute strategies to mitigate their potential health and performance implications. Indeed, there is evidence of favorable outcomes when such an approach is implemented (26,28,45). Nevertheless, there is renewed focus on health and performance effects of chronic energy deficits in athletes, because of the recognition of concerns associated with low energy availability and the sequelae described in the Female and Male Athlete Triad (46) the Relative Energy Deficiency

Table 1.
Summary of weight-category sports and their respective weigh-in characteristics.

	Weight Categories (9)	Weigh-in Procedures	Competition Characteristics	Comments
Combat Sports				
Boxing (amateur/"Olympic Style") (10)	AIBA competition: M = 10, F = 10 Olympic Games: M = 8, F = 5	General weigh-in on the first day of tournament, and on the morning of the day in which specific competition is drawn. Weigh-in, is no less than 3 h before 1 st bout of the day. Official weigh-in can only be attempted once, with athletes allowed to wear a swimming suit or undergarments. When necessary, the weigh-in, can be conducted naked.	Tournament contested over different days with 4 to 5 bouts in competition, spread every 2nd day. M: 3 × 3 min rounds F: 3 × 3 min rounds	Different weight categories across competitions (e.g., Olympic Games vs others): some athletes must change to a new weight category.
Boxing (Professional) Note that there are different world bodies governing professional boxing: WBA (11) is provided as an example	Varies by country/association WBA: M = 17, F = 16	Procedures vary between associations. The official scale is made available to check weigh > 2 h prior to official weigh-in. WBA title fights: official weigh-in = > 16 h and < 30 h before start of the first bout on the event card. If either champion or challenger fails to make prescribed weight, each will have further 2 h	Title fights: 12 rounds Other fights may vary between 4 and 12 rounds. M: 3 min rounds F: 2 min rounds Competitors usually fight 3 to 4 times per year	Boxers may change weight categories to seek titles. Often lengthy periods between fights, but fighters may not prepare until bout is contracted, thus creating short preparation. Culture often leads to large swings in weight between fights. Failing to make weight can result in imposing a monetary fine, and demotion or suspension.
Judo (12)	All competitions M = 7, F = 7	An alternate scale is made available in the training room for days prior to the official weigh-in, which occurs the evening before competition. Random weight checks morning of competition prohibits weight gain of > 5% above weight category. Official weigh-in can only be attempted once, with athletes allowed to wear undergarments. When necessary, the weigh-in can be conducted naked. Cadet athletes are given + 0.1 kg allowance for underwear.	Tournament contested on a single day: 4 to 8 bouts. M: 1 × 4 min round F: 1 × 4 min round	Same weight categories across Olympic Games and all competition. Random reweigh on fight day may discourage recovery strategies
Karate (13)	Senior competition: M = 5, F = 5 Olympic Games: M = 3, F = 3	The official scale is made available to check weight > 1 h prior to official weigh-in, which must take place, at the latest, the day before the start of competition, unless specified otherwise. All athletes are given a + 0.2 kg allowance as official weigh-in must be conducted in undergarments (M: underpants F: underpants and bra).	Tournament contested on a single day: 6 to 8 bouts. M: 1 × 3 min round F: 1 × 3 min round	Different weight categories across competitions (e.g., Olympic Games vs others): some athletes must change to a new weight category

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	Weight Categories (9)	Weigh-in Procedures	Competition Characteristics	Comments
Mixed Martial Arts Note that there are different world bodies governing Mixed Martial Arts: UFC (14) is provided as an example	Unified Weight Classes – MMA 14 classes UFC Titles M = 8 F = 4	An alternate scale is made available in the training room for days prior to the official weigh-in, which occurs on the morning of the day before the fight (26 to 32 h preevent). A public “staged” weigh-in is held on the afternoon of the day before fight. Official weigh-in can be attempted multiple times, with athletes allowed to wear undergarments. When necessary, the weigh-in can be conducted naked.	Major “fight nights” each month but fighters normally engage in two to three fights per year. Fight duration varies between title and exhibition fights Title or main event: 5 × 5 min rounds Other events: 3 × 5 min rounds	MMA athletes may change weight categories to seek titles. Often lengthy periods between fights, but fighters may not prepare until bout is contracted, thus creating short preparation. Culture often leads to large swings in weight between fights. Weigh-ins for title matches are televised. Allowances (and penalties of up to 30% of purse) sometimes made for fighters over weight.
Taekwondo (15)	World championships: M = 8; F = 8 Olympic Games: M = 4; F = 4	An alternate scale is made available in the training room for days prior to the official weigh-in, which occurs on the day before competition. Random weight checks the morning of competition prohibit weight gain of > 5% above weight category. Official weigh-in can only be attempted twice, with athletes allowed to wear undergarments. When necessary, the weigh-in can be conducted naked. Cadet and Junior athletes are given +0.1 kg allowance for underwear.	Tournament contested on a single day: four to eight bouts. M: 3 × 2 min rounds F: 3 × 2 min rounds	Different weight categories across competitions (e.g., Olympic Games vs others): some athletes must change to a new weight category Random reweigh on fight day may discourage recovery strategies
Wrestling (Freestyle) (16)	UWW competition: M = 10; F = 10 Olympic Games: M = 6; F = 6	Weigh-in on the morning before competition. There is another weigh-in on the second day of competition with no weight allowance (2 kg is tolerated at world cups and international tournaments). After passing a medical check, athletes must wear a competition singlet during weigh-in.	Tournament contested across 2 d: four to eight bouts. M: 2 × 3 min rounds F: 2 × 3 min rounds	Different weight categories across competitions (e.g., Olympic Games vs others): some wrestlers must change to a new weight category
Wrestling (Greco Roman) (16)	UWW competition: M = 10 Olympic Games: M = 6	Weigh-in on morning before competition. There is another weigh-in on the second day of competition with no weight allowance (2 kg is tolerated at world cups and international tournaments). After passing a medical check, athletes must wear a competition singlet during weigh-in.	Tournament contested across 2 d: four to eight bouts. M: 2 × 3 min rounds F: 2 × 3 min rounds	Different weight categories across competitions (e.g., Olympic Games vs others): some wrestlers must change to a new weight category
Collegiate Wrestling (17)	NCAA and NCWA: M = 10; F = 8	Weigh-in on the day of competition no less than 2 h before 1 st bout. Athletes must wear undergarments at weigh-in and follow a regulated certification program inclusive of < 1.5% weekly BM loss, minimum 5% body fat, etc. prior to competition.	Dual meets: single day competition between teams from 2 (or more) colleges with single match for each weight category. Tournament: multi-day meet with 2 matches per day. 7 min bouts (3 + 2 + 2 min rounds).	Weekly dual/multiple meets over competition season with tournaments at end of season. Certification program and other protocols attempt to limit unsafe weight loss practices.

Other Sports		
Olympic Weightlifting (18)	IWF: M = 10; F = 10 Olympic Games: M = 7; F = 7	Weigh-in on the day of competition no less than 2 h before 1st event of the day. Official weigh-in can be attempted multiple times, with athletes allowed to wear undergarments. When necessary, the weigh-in can be conducted naked.
		Tournament contested on a single day. Three lifts in two disciplines (clean-and-jerk and snatch). In case of tie, lightest competitor wins.
		Different weight categories across competitions (e.g., Olympic Games vs others): some lifters must change to a new weight category
Powerlifting (19)	IPF: M = 8; F = 8	Weigh-in on the day of competition no less than 2 h before 1st event of the day. Official weigh-in can only be attempted once (reweigh at discretion of referees), with athletes allowed to wear undergarments and socks. When necessary, the weigh-in can be conducted naked.
		Tournament contested on a single day.
		Three lifts in three disciplines (squat, bench press, and deadlift)
Rowing (20) (lightweight)	M: Average crew weight < 70 kg No individual > 72.5 kg F < 57 kg, 59 kg Coxswain: > 55 kg (M) and > 50 kg (F/mixed)	Regatta over ~ 7 d Most crews race every second day Race = 2000 m course
		Many lightweight rowers only race "at weight" a couple of times per year at important races. Many countries provide opportunities in early season races for lightweight rowers to race above weight. Crew boats may require some rowers to be lighter to provide opportunity for heavier rower within boat average
American "Sprint" Football Collegiate League (21)	M: < 170 lb (77 kg) or < 183 lb (83 kg) with > 5% BF and USG ≤ 1.020	Competitive season involves maximum of seven formal games, and more recently, a play off between top teams of the two divisions.
		Complies with American Football rules but promotes speed and agility rather than mass/strength.
Horse racing (jockey)	Different weight handicaps between types of races (e.g., flat vs steeple) and different countries or local race authorities	1 to 2 race meets a week over racing season. Jockeys may have six to eight rides during one race meet over 5 to 6 h. Races are conducted over various distances, usually 1000 to 2000 m (lasting 1 to 2 min), but occasionally longer.
		Multiple races over a single race meet may require different weight handicaps according to the horse. Need for weigh-in after race prevents recovery strategies.

M, male; F, female; AIBA, International Boxing Association; UWW, United World Wrestling; UFC, Ultimate Fighting Championship; NCAA, National Collegiate Athletic Association; NCWA, National College Wrestling Association; IWF, International Weightlifting Federation; IPF, International Powerlifting Federation.

in Sport (RED-S) syndrome (47), including specific issues in weight category sports (48). Since this enhanced interest is beyond the scope of this Statement, the reader is directed to specific resources on strategies and concerns for chronic weight loss in athletes (49–53), allowing the current article to focus on RWL while acknowledging that it occurs against a larger backdrop of body composition manipulation.

Table 2 provides a summary of commonly utilized RWL strategies, including their associated potential benefits and disadvantages. It is important to understand the degree to which these strategies are utilized when attempting to infer health and/or performance implications. Acute restriction of fluid intake and/or promotion of sweat loss through exercise to levels commonly observed in routine training (e.g., 2% to 3% BM reduction) are likely to have few disadvantages, especially if aggressive recovery strategies are implemented subsequent to weigh-in. However, a singular focus on these practices to facilitate larger amounts of weight loss (>3% BM reduction) may have substantial negative implications to health and performance. According to recent surveys of RWL practices among athletes, the concerning strategies of vomiting and use of banned diuretics and diet pills have a low prevalence of use (3,24,25). However, the use of thermally stressful environments like saunas to elicit sweat losses remains common (3,24,25). There also appears to be an increased interest in the use of other novel strategies such as water loading (3,24,25), a practice in which the health and performance implications are only now beginning to emerge (55). The continued reliance on RWL strategies for weight making should not come as a surprise, since athletes identify fellow athletes and coaches as primary influencers on their BM management practices (2,24,41). It also should be noted that parents have an important influence on young athletes, but lesser so as athletes age (56,57). Practitioners who work with weight category athletes should have a clear understanding of their specific weight making practices and timeframes of implementation, not only to assist with better management of weight loss goals, but to consider postweigh in recovery practices that can mitigate the health and/or performance consequences.

Opportunities for Recovery after Weigh-in

The period between weigh-in and the event offers an opportunity for an athlete to consume foods and drinks that address preevent nutrition goals against a background of the physiological and psychological challenges imposed by their RWL practices (54). What is needed to optimize the athlete's nutritional status for competition will vary according to the type and magnitude of RWL practices undertaken and the competitive event. What is practical to achieve will depend on whether the timeframe will allow consumption, digestion/absorption, and restoration of suboptimal nutritional status (54). Table 3 summarizes common sports nutrition advice for such processes, emphasizing restoration of body fluids and muscle glycogen stores as common goals (58,59), and noting specific recommendations should be applied with caution and practical considerations. For example, it may not always be possible to fully recover in the available time period; here the athlete should prioritize recovery of what is most important for their event and ensure that recovery eating does not interfere with event performance, either by creating gut discomfort from the consumption of large volumes of fluid and nutrients or interfering with the requirement to make weight again within day(s).

Several studies have examined how well recovery can be achieved in the typical timeframe between weigh-in and event, investigating the change in physiological parameter (e.g., degree of dehydration) and/or the restoration of performance. The literature is difficult to fully interpret since studies differ in the magnitude of weight loss, whether single or several strategies have been involved in the weight making and/or recovery, and the time course of each process. Most importantly, studies are typically laboratory based, quantifying performance in measures of work, power, or force production. Although these measures are presumed to contribute to success in competition, they may not be a perfect proxy. Furthermore, they may not consider that competitive success is a relative outcome (e.g., comparing athletes to each other) rather than absolute (comparing an athlete to their personal best). The results of studies that have measured the effects of recovery strategies on performance in a meaningful way are discussed in the following section.

Alternative methods of recovery, such as intravenous fluid administration, might have appeal following the weigh in. Studies demonstrate some efficacy in nonweight-category athletes who partially rehydrate by this means but the effects are no better than those of orally administered fluids (60,61). Use of such methods are strongly discouraged given they are banned by sports governing bodies.

Concerns Associated with Weight Making Practices

Weight making strategies may jeopardize the athlete's health and safety and alter attributes or whole elements of performance. Since the issues of health and safety should be of primary concern, these will be addressed first. The most serious consequence of making weight is death; such outcomes have been reported in association with extreme practices in taekwondo (62), college wrestling (63), MMA (64,65), judo (66), Muay Thai (67), and lightweight rowing (68). Such tragic and avoidable occurrences are typically associated with cardiovascular complications and/or heat stroke secondary to severe fluid restriction, passive heat exposure or strenuous exercise to induce large sweat losses, or combinations of these strategies (62–68). Indeed, substantial reduction in body water, including plasma volume, is associated with cardiovascular strain and impaired thermoregulation in hot environments (69,70). Depending on the method of water loss, electrolyte imbalances also may occur, and, according to altered electromyography activity, increase the risk of muscle heat cramps (71). Acute renal stress or injury also has been associated with extreme fluid loss during RWL in some weight category athletes (27,72). Although the concept of RWL prior to a competitive event suggests that it is a short-term experience, it should be noted that in sports involving regular competition (e.g., weekly matches during the college wrestling season), this protocol is repeated many times.

Chronic exposure to negative energy balance and low energy availability is associated with effects on many aspects of health and function including impairments to metabolic, endocrine, and reproductive systems, as well as protein synthesis and growth in child and adolescent athletes; all contribute to the syndrome known as RED-S (47). Reductions in testosterone and insulin-like growth factor 1 concentrations have been observed during the 3- to 4-month competitive wrestling season (27,28,73–75), as well as in lightweight rowers (76) and jockeys (77). Acute markers of impairments in protein status

Table 2.**Benefits and disadvantages of common methods of rapid weight loss (modified from 54).**

Acute Weight Loss Method		Benefits	Disadvantages
Body water manipulation	Passive sweating (thermally stressful conditions <i>e.g.</i> , sauna, hot bath, sweat suits etc.)	• Relatively simple method of weight loss • Promotes high sweat rates and thus weight loss	• Preferential loss of plasma volume • Thermally stressful environment exacerbates risk of hyperthermia • Additional exercise may induce fatigue/muscle soreness
	Active sweating (exercise induced)	• Can be easily incorporated into existing training sessions before weigh-in	
	Fluid restriction	• Better maintains plasma volume (than passive sweating)	• Additional exercise further reduces muscle glycogen, requiring more aggressive refueling recovery • Increased thirst sensation • Slower rate of loss than promotion of sweat rate • Still experimental and likely to achieve smaller net fluid (BM) loss than other techniques
	Water loading (intake of large volumes of fluid for days, followed by sudden fluid restriction on the day before weigh-in)	• May cause less physiological disturbances than other forms of dehydration	
	Diuretics	• Relatively simple to implement and passive	• Small risk of hyponatremia • Banned in most sports • Principally reduces plasma volume • Electrolyte imbalance and risk of muscle cramps • Likely banned in most sports • Requires sterile conditions to mitigate infection risk
	Blood withdrawal preweigh-in (for reinfusion post weigh-in)	• May facilitate greater satiety despite low energy intake • Large volume of urine loss with no energy expended • Moderate volume of fluid removed without need for energy expenditure	
Manipulation of body energy stores (principally achieved in the short-term via a reduction in body carbohydrate stores and associated water)	Exercise	• Can be integrated into training program	• Glycogen replacement required for performance needs in many events • If exercise is additional and excessive, can cause fatigue/muscle soreness • Hunger, loss of perceived energy, and reduction of muscle glycogen • Restriction of dietary carbohydrate associated with fatigue (especially during exercise)
	Restriction of energy intake Specific reduction of dietary carbohydrate intake	• Can be achieved by change in energy density of food choices • Allows continuation of food intake (<i>i.e.</i>, absence of hunger) while still achieving the reduction in muscle glycogen/water content	
Manipulation of gastrointestinal tract contents	Food restriction	• Easily implemented	• Hunger, loss of perceived energy, and reduction of muscle glycogen • Only small impact on body mass • Banned in some sports • Body water loss and associated impaired cardiovascular function • Electrolyte imbalances
	Fiber restriction Laxative/bowel preparation use	• Minimal impact on acute nutritional status or performance	

in these wrestlers include decreases in rapid-turnover plasma proteins indicative of protein-calorie nutrition status (73,78), while rates of muscle protein synthesis in response to resistance training are reduced with several days of low energy availability (LEA) (79). Meanwhile, its long-term translation into loss of FFM has been reported across a wrestling season (73,78) and as a significant contribution to the BM losses in a boxing training camp (26). Reduced bone mineral density is often associated with LEA but may be altered by the interaction with bone loading exercise. For example, although some studies of combat athletes have shown evidence of adverse outcomes for acute markers of bone turnover, bone mineral density (BMD) did not appear to be affected by weight reduction (28,80,81). However, lightweight rowers report lower BMD than their heavy weight counterparts and a high risk of rib stress fractures (48). Conflicting outcomes exist within the same sport with low BMD reported in jockeys (23) but no differences in BMD observed between novice and veteran jockeys who have many more years of experience and exposure to making weight (82). Other outcomes associated with RED-S observed

among weight category athletes include dyslipidemias and reduction in resting metabolic rate (48). The effect of LEA on the immune system is complex (83), and although some markers of diminished immune system function have been reported in weight category athletes (84–89), it is unclear whether these changes put weight-category athletes at greater risk for illness during heavy training or weight loss periods.

The effects of RWL, or its components, on behavioral and psychological issues have been considered from a number of angles. With potential pressure to either lose weight and/or to maintain meticulous control of body composition, it is not surprising that athletes may exhibit disordered eating or eating disorders. Studies have reported that some but not all cohorts involved in weight making sports have higher scores on eating disorder survey instruments than age-matched controls or other athletes (23,90,91). Other investigations have focused on changes in measures of mood states during RWL activities, with consistent findings of reductions in vigor, and elevations in anxiety, anger, fatigue, depression, and confusion (23,92,93). Transient and reversible impairment of cognitive

Table 3.**Strategies for recovery of fluid and carbohydrate stores between weigh-in and event.**

Issues	General Recommendations	Practical Issues to Consider in Weight Category Sports
Rehydration (58)	• Consume a fluid volume equal to ~ 150% of the fluid deficit to allow for ongoing urine and sweat losses during the recovery period • Replace lost electrolytes, particularly sodium, as part of the fluid plan to promote rapid reequilibration of fluid compartments and to minimize urine losses. This can be achieved by consuming electrolyte-containing drinks that are specifically developed for rehydration (<i>e.g.</i>, oral rehydration solutions or ~ 50 to 60 mmol sodium) and/or consuming salt-rich foods with other drink choices • Note that the addition of protein to a drink or in food consumed at the same time can enhance effective rehydration by enhancing fluid retention, and may contribute to other recovery goals • The rate and volume of intake of fluid needs to consider gastric emptying (larger volumes increase rate of emptying) vs fluid retention (a slower more frequent intake of fluid will reduce perturbation in plasma osmolality and reduce urine losses)	• It may not be possible to consume enough fluid in the available timeframe between weigh-in and event to allow for full restoration of fluid losses; a protocol that integrates considerations around gastro-intestinal absorption rates, urinary losses, and gut comfort for the event should be developed according to individual experience • Replacement of electrolytes is more important when dehydration has been achieved via sweating (with accompanying loss of electrolytes) than fluid restriction • Consuming a large fluid bolus (<i>e.g.</i>, 10 mL·kg⁻¹) immediately after weigh-in, followed by continued smaller volumes at regular intervals will maintain a high rate of gastric emptying (via high gastric volume). This also may reduce the need to consume fluids just before the event, risking gut discomfort during competition • If further weigh-ins are required (<i>e.g.</i>, multiple days of competition), the athlete should rehydrate for the first bout according to the importance of hydration to performance, then repeat the dehydration-rehydration protocol for subsequent days • If further weight limits are imposed (<i>e.g.</i>, mat-side weigh-in with limits for regain following official weigh-in), the athlete should be cautious that initial weight making was not so severe that appropriate competition preparation cannot be achieved with the allowable recovery regain
Glycogen storage/restoration of CHO availability (59)	• Take advantage of enhanced rates of glycogen restoration when muscle stores are most depleted and in the period (up to 4 h) immediately after an exercise bout. Adequate energy availability is required to optimize glycogen storage from a given amount of CHO. • Enhance early postexercise recovery with a higher rate of CHO intake (~ 1 g·kg BM⁻¹·h⁻¹), especially by consuming in frequent small feedings. • Focus on CHO-rich foods with a moderate to high GI to provide a readily available source of substrate for glycogen synthesis. • Add protein to CHO-rich meals and snacks to promote glycogen storage when CHO intake is suboptimal, especially during the first hours of recovery. An intake of ~ 20 g to 25 g of high-quality protein appears to optimize this effect while also meeting goals for postexercise muscle protein synthesis. • Target daily CHO intake based on body mass (or proxy for the volume of active muscle) and exercise load. Guidelines can be suggested but need to be fine-tuned according to the athlete's overall dietary goals and feedback from training. - o Moderate exercise load: 5 to 7 g·kg BM⁻¹·h⁻¹ - o Heavy exercise load to glycogen supercompensation: 6 to 1 g·kg BM⁻¹·h⁻¹	• Although the athlete may have reduced glycogen stores as a result of weight making strategies, the importance of glycogen to performance will vary according to the type of sport and the number of events/bouts that will be undertaken on a day. The athlete should refuel accordingly • It may not be possible to refuel sufficiently if the postweigh in period is brief (<i>e.g.</i>, 1 h to 2 h) and weight making practices have involved severe CHO restriction and/or exercise. The athlete should consider this when undertaking weight making practices. • Food choices should consider CHO targets, other nutritional goals (<i>e.g.</i>, protein intake), and gut comfort. Compact and fluid-based choices may be most suitable, and intake of fat and fiber should be moderated.

CHO, carbohydrate; BM, body mass.

functioning, such as memory and ability to complete simple tasks, has been reported in some athletes during active weight making (94,95), with potential effects on performance outcomes and injury. However, it is important to also consider other, including perceived positive, effects of the psychosocial aspects of RWL. For example, a survey of the attitudes of elite combat

athletes reported that their weight regulation practices provided them with a sense of identity as an athlete, contributed to a sense of focus and commitment to precompetition preparation, and allowed a feeling of advantage over an opponent (96,97).

Focusing specifically the effects of acute dehydration on cognitive function, magnetic resonance imaging studies have revealed

Table 4.

Effects of weight reduction on performance and success of weight-category athletes*.

Intervention Study: Chronic Weight-Making without Recovery Strategy				
Sport	Strategy of Weight Making	Magnitude of BM Loss (%BM Mean)	Measurement of Performance	Outcome with Weight Making (versus Baseline or Control Group)
MMA (27)	Case study; M = 1: 7-wk energy restriction + 5 d of water loading	18.1%	• $\dot{V}O_2$ peak on treadmill	Inability to complete $\dot{V}O_2$ peak test ($\downarrow$ 11.1% in $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$).
MMA (28)	Case study; M = 1: 8-wk energy restriction (7 wk at RMR and 5 d further reduced)	13.5%	• Upper and lower body maximum dynamic strength (1RM), • Maximum dynamic power (MDP) • $\dot{V}O_2$ peak on treadmill	$\uparrow$ strength: 6% to 9% absolute, 18% to 19% relative $\uparrow$ MDP: absolute upper body, relative lower body $\uparrow$ 13% absolute $\dot{V}O_2$ peak $\uparrow$ 19% relative $\dot{V}O_2$ peak
Intervention study: Rapid Weight Loss with Minimal or no Recovery Opportunity				
Sport	Strategy of Weight Making	Magnitude of BM Loss (%BM Mean)	Measurement of Performance	Outcome with Weight Making (versus Baseline or Control Group)
Boxing (106)	5-d training and energy/fluid restriction; M = 8	3.0%	• Punching force during simulated boxing bout (3, 3-min rounds)	$\downarrow$ 4.6% (NS) in punching forces ($P > 0.05$)
Boxing (107)	~2-h of low-intensity exercise in a sauna and sweat suit; M = 7	3.8%	• Punching force during simulated boxing bout (3, 3-min rounds with 108 punches per round)	No change in punching force ($P > 0.05$)
Boxing (108)	Athlete-preferred methods using training and energy/fluid restriction, n = 16 (sex NA)	5.2%	• Number of repetitions performed during intervals of efforts consistent with boxing training	$\downarrow$ ~6.7% in repetitions compared to set goals ($P < 0.05$)
Judoists (109)	7-d food restriction; M = 11	4.9%	• Isometric grip strength • Vertical jump flight time and number in 7 s and 30 s • Single standing jump (SJ) and counter-movement jump (CMJ)	Significant $\downarrow$: left grip and jump performance for 30 s ($P < 0.05$) NS: right grip, 7 s jump number, SJ, or CMJ ($P > 0.05$)
Judoists (110)	Gradual (3 wk) reduction with food/fluid restriction plus rapid reduction with exercise in sweat suits (dehydration) vs baseline; M = 5, F = 5	4.0%	• Judo movements in 5 s or 30 s • Vertical jumps: squat jumps and CMJ	$\downarrow$ 10% for of judo movement repetitions over 30 s ($P < 0.05$) NS for 5 s movements or either vertical jumps ($P > 0.05$)
Judoists (111)	~1 month self-prescribed WL vs those who maintained or gained weight, M = 27	1.1%	• Forearm maximum isometric contraction (MVC) with visual feedback.	NS: MVC overall Those with $> 2\%$ $\downarrow$ MVC had greater weight reduction ($P < 0.05$) Those with $\downarrow$ TBW had $\downarrow$ MVC ($P < 0.05$)
Combat athletes (112)	RWL over 5 to 7 d via food/fluid restriction and sweating; and post competition (nondescript recovery) M = 14 RWL, M = 14 controls	5.5% in RWL group	• $\dot{V}O_2$ peak and peak running velocity • Running speed at which 4 mmol lactate $\cdot \text{L}^{-1}$ (4LT)	No change in $\dot{V}O_2$ peak or peak velocity with RWL ($P > 0.05$) $\uparrow$ running speed at 4LT

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Sport	Strategy of Weight Making	Magnitude of BM Loss (%BM Mean)	Measurement of Performance	Outcome with Weight Making (versus Baseline or Control Group)
Wrestler — collegiate (93)	4-d energy restriction but iso-protein with high CHO vs low CHO content; M = 12	6.2%	• Intermittent high-intensity isokinetic arm ergometry (total work performed)	NS differences in work on high-CHO diet ($\downarrow 0.8\%$, $P > 0.05$); $\downarrow$ work on low-CHO diet ($\downarrow 8.7\%$, $P < 0.05$)
Wrestling — collegiate (113)	10-d self-prescribed weight loss of different magnitude (categorized by authors); M = 16	$\sim 1\% \sim 3\% \sim 6\%$	• Maximum isometric grip strength • Leg ergometer Wingate test	No change in performances before and after weight reduction regardless of category % weight loss ($P > 0.05$)
Intervention study: Rapid Weight Loss with Recovery Opportunity				
Sport	Strategy of Weight Making and Recovery	Magnitude of BM Loss (%BM Mean)	Measurement of Performance	Outcome
Combat athletes (judo, jiu-jitsu, MMA and wrestling) (114)	Self-prescribed WM in weight cyclers, M = 10, vs nonweight cyclist athlete, M = 8, with 4-h recovery of approximately 1,600 Cal, 250 g CHO, 50 g pro, 45 g fat, and ad-lib water (volume not specified)	5%	Intermittent high-intensity isotonic arm ergometry work	NS performance difference: weight cyclers vs noncyclers ($P > 0.05$) NS composition of recovery meal: weight cyclers vs noncyclers ($P > 0.05$)
Judoists (115)	Self-prescribed training and food and fluid restriction, M = 10, vs nonweight loss control group, M = 10	3%	Five judo matches, each followed by • Maximum isometric hand grip strength anaerobic capacity test of upper body (rowing)	$\downarrow$ in performance with cumulative matches in WM group
Judoists (116)	5-d self-prescribed WM + 4-h ad-lib fluid and food recovery providing 1391 Cal, 201 g, 50 g fat, 34 g protein; M = 7 vs control group, M = 7	4.8%	• 3- min high-intensity judo drills against an opponent + 5-min rest • 5-min judo combat against comparable opponent (scoring of activity time, number of attacks and attempts) + • 15 min recovery • Three $\times$ 30-s Wingate tests with 3-min recovery	NS difference in activity time, number of attacks and Wingate performance ($P > 0.05$).
Lightweight rowing (M) (117)	24-h food + fluid restriction + exercise-induced sweating in sweat suit Recovery post weigh-in (2 h): 1500 ml water; M = 8	5.2% (before recovery)	• Rowing ergometer TT simulating 2000 m	Performance $\downarrow$ 22 s ($P < 0.05$)
Lightweight rowing (M + F) (118)	24-h increased exercise and reduced food/fluid intake + 2 h recovery: 2.3 g·kg ⁻¹ CHO, 25 mL·kg ⁻¹ fluid + 34 mg·kg ⁻¹ Na, and ad lib fluids; M = 8, F = 9	4% (before recovery)	• 2000 m rowing ergometer TT	Performance $\downarrow$ 2.1 s ($P = 0.003$) or 0.7%
Lightweight rowing (M + F) (33)	24-h increased exercise and reduced food/fluid intake + 2 h recovery: 2.3 g·kg ⁻¹ CHO, 25 mL·kg ⁻¹ fluid + 34 mg·kg ⁻¹ Na and ad lib fluids; M = 8, F = 9	4% (before recovery)	• 1800 m on water rowing TT (cool conditions)	Performance $\downarrow$ 1.0 s ($P = 0.29$)

Lightweight rowing (M) (34)	24-h increased exercise and reduced food/fluid intake + 2 h recovery: $2.3 \text{ g} \cdot \text{kg}^{-1}$ CHO, $28 \text{ mL} \cdot \text{kg}^{-1}$ fluid + $34 \text{ mg} \cdot \text{kg}^{-1}$ Na and ad lib fluids then 2 further trials 48 h apart	4% (before recovery)	• $4 \times 2000 \text{ m}$ rowing ergometer TT, 48 h apart, (simulates multi-day regatta where weight must be made on any race day)	When outliers removed: Performance $\downarrow 3.0 \text{ s}$ ($P = 0.07$) for first WM race Performance regained in subsequent races, especially with full recovery
Combat athletes (119)	• Complete Recovery posttrace (regain of 3%) + full WM; M = 6 OR • Partial Recovery posttrace (regain of 0.3%) + reduced WM; M = 6 • No weight loss control; M = 4	3.2%	• Maximum voluntary isometric knee extension (MVC) • Rate of force production, indices of neural recruitment, • Total number repetitions at 85% MVC	• $\downarrow 26.1\%$ repeated contractions at 85% MVC ($P < 0.05$) • Core temp elevated to 39°C with RWL ($P < 0.05$) • All other outcomes NS ($P > 0.05$)
Taekwondo (120)	3.5 d self-prescribed training, passive and active sweating, energy restriction prior to official weigh with performance done after 16 h recovery using self-prescribed approach of rehydration and carbohydrate intake; M = 5	5.0%	Three simulated TKD matches quantifying • kicking frequency • attack time & number • blood lactate • $\dot{V}\text{O}_2$	• $\uparrow$ kicking frequency, $\dot{V}\text{O}_2$ (peak and mean), and lactate for those who lost weight ($P < 0.05$)
Taekwondo (121)	2-w sport-specific training, passive & active sweating, energy/fluid restriction prior to official weigh and 24-h period (no details) before performance tests; M = 15 vs nonweight loss control group, M = 16	10.3%	Three X 2-min bouts of TKD against comparable opponent • matches scored for skill performance including scoring, action and no action.	• $\uparrow$ performance scores in control group vs no change in scores for weight-loss group ($P < 0.05$).
Wrestling — collegiate (122)	3 to 5 d self-prescribed weight loss then 1 to 2 h recovery with $\sim 1 \text{ L}$ water + 1.5 g kg CHO or $\sim 1 \text{ L}$ flavored placebo; M = 15	4.6%	• Intermittent high-intensity isotonic arm ergometry	NS change in arm work with weight loss or recovery ($P > 0.05$)
Wrestling — collegiate (123)	3 to 5 d energy-restricted standardized high CHO liquid diet and training, then 5-h recovery with high CHO (75%; $21 \text{ kcal} \cdot \text{kg}^{-1}$) or isocaloric mod-CHO (47%, $21 \text{ kcal} \cdot \text{kg}^{-1}$); M = 12	3.3%	• Intermittent high-intensity isotonic arm ergometry	• $\downarrow 7.6\%$ arm work after weight loss ($P < 0.05$) and trend for no change ($\downarrow 0.9\%$) work after recovery on high-CHO treatment vs $\downarrow 8.5\%$ work on mod-CHO treatment ($P = 0.10$)
Observational study: Performance Outcomes in Real-life Events using Magnitude of RWG as Proxy for Magnitude of RWL				
Sport	Reweight Information	Magnitude of RWG (%BM Mean)	Measurement of Performance	Outcome
Boxing - professional boxing (M) (124)	RWG between 24 h (weigh in) and 12 h (2nd weigh) preflight; M = 126, F = 16	3.8%	• 71 IBF fights	Negligible correlation between fight outcome and RWG ($P > 0.05$)

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Sport	Strategy of Weight Making	Magnitude of BM Loss (%BM Mean)	Measurement of Performance	Outcome with Weight Making (versus Baseline or Control Group)
Boxing — Olympic-style elite adolescent (125)	Survey on weight reduction practices in European boxers; M = 83	2.8%	• Success in European Championships (medal vs nonmedal)	RWG not associated with success (odds ratio; 1.70, $P > 0.05$). Boxing experience linked to success (odds ratio: 1.33, $P < 0.05$)
Boxing — Olympic style senior M & F (38)	RWG during 3 to 12 h between official weigh in and 1 h before fight; M = 70, F = 30	2.1% for M; 1.5% for F	• Finalists vs nonfinalists and winners vs losers of matches	No difference in RWG between winners vs losers and finalists vs nonfinalists ($P > 0.05$)
MMA (126)	Typical methods of RWL over 15 to 30 d and RWG 24 h between weigh in and match M = 8 bout winners, M = 7 bout losers	RWL ~ 7% RWG: 3% losers 6% winners	Match outcome — win vs loss • Technical aspects of match	No difference in RWL for winner vs loser ($P > 0.05$) RWG greater for winners ($P < 0.05$) Technical-tactical and time-motion performance associated with RWG
MMA professionals (40)	24-h recovery between weigh in and bout. 31 winners, 31 losers Sex not stated	RWG: 9.1% losers 10.1% winners	• Match outcome — win vs loss	No association between RWG ($P > 0.05$); both winners and losers regained substantial weight
Wrestling — collegiate (127)	RWG between 17 h official weigh in and 1 h before match; M = 607	5%	• Success in 1 st round of US collegiate national championships	No relationship between success and RWG or weight discrepancy between 1 st round opponents ($P > 0.05$). No difference in RWG for place winners (5.2%) vs nonplacers (5.3%) ($P > 0.05$)
Wrestling — adolescent (128)	12 h between official weigh in and check-in before match; M = 260	2.2%	• Victory in the 1 st round of day 2 of a tournament	1 st round winners regained more weight (2.4%) than 1 st round losers (1.9%) ($P < 0.05$)
Judoists — senior, M and F (36)	RWG during 15 to 20 h between official weigh in and 1 h before first fight; F = 36, M = 50	2.3% for M and 3.1% for F	• Medalists vs nonmedalists and winners vs losers of matches	Medalists had greater RWG than nonmedalists, especially for males ($P < 0.05$); Winners had greater RWG than losers of matches ($P < 0.05$)
Observational study: Performance Outcomes in Real-life Events with Weight Loss Reported				
Sport	Strategy of Weight Making	Magnitude of BM Loss (%BM Mean)	Measurement of Performance	Outcome
MMA (129)	30-d fluid/energy/CHO restriction plus 24-h passive and active sweating; M = 12	9.4%	• Victory in fights	No clear relationship between RWL and success
MMA (39)	7-d self-prescribed weight loss before official weigh in. 24-h preevent recovery not described; M = 73, F = 2	8.6% and 10.6%	• Victory in fights	Greater RWL (10.6%) in those who lost fight than those who won (8.6%)
Observational Study: Effects of a Season of Weight Loss				

Wrestling — adolescent (130)	Typical wrestling training and diet restriction; M = 9, vs maturation- and size-matched physical active controls; M = 7. Measurements at pre and late competitive season (3.5 to 4 month span of repeatedly making weight).	3.8%	• Strength assessed using isokinetic peak torque at elbow and knee at 60°/s and 180°/s ↓ strength for all assessments in association with ↓FFM (2%) ($P < 0.05$). In contrast, for physically active controls, ↔ strength ($P > 0.05$) and ↑FFM (3.5%) ($P < 0.05$).
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CHO, carbohydrate; MVC, maximum voluntary contraction; NA, not available; Na, sodium; NS, not significant statistically; TT, time trial.

*The literature summarized here was obtained using PubMed searches on the following combination of phrases: boxing, combat sports, judo, mixed martial arts, taekwondo, or wrestling *and* dehydration, making weight, weight loss, or weight reduction, *and* athletic performance, performance, or physical performance. Reporting of change in body mass was a requirement for inclusion in the summary. Acceptable publication dates of the articles started with 1996 (date of ACSM position stand on weight loss in wrestlers) and continued to the present. All published observational studies on weight manipulation and sport-specific success were considered if not included. Intervention studies that were deemed acceptable compared physical performance at baseline vs postweigh-in or postnutrition recovery following weight loss practices typical of the sport-specific combat athletes at the date of publication. Intervention is defined as protocols involving weight manipulation for research purposes, not necessarily for sport competition. Weigh-in protocols change over eras and are sport-specific, but the summary is broad enough to allow a general but valid conclusion regarding advantages and disadvantages of various weight loss scenarios among athletes in weight-category sports.

temporary alterations in the brain structure (98–100), which have been associated with transient declines in cognitive test scores (101) and visuomotor performance (102). Interestingly, wrestlers performed worse on concussion screening tests when dehydrated compared with their performance at baseline in an euhydrated state (103). Wide variability in concussion testing responses in nonconcussed boxers also has been explained in part by dehydration (104). Implications of structural changes or possibly reduced brain blood flow with dehydration (105) include that the brain might be more physically vulnerable to concussion, or that impaired decision making and motor control during competition may predispose weight-restricted athletes to injuries. However, to date, empirical data are lacking to confirm an increased risk of brain injury with dehydration.

It is important to consider the real and perceived effects of RWL on performance in weight category sports since this is likely to underpin the motives of athletes and coaches around their practices and the capacity for change. A summary of studies of the effect of RWL on sport/exercise performance (Table 4) illustrates the complexity of this issue by demonstrating a range of outcomes (trivial effects, negative effects, and positive effects) from a range of study designs. Indeed, no single answer is likely because, as demonstrated in Table 1, chronic weight loss, RWL practices, and the characteristics of performance are all variable. Studies of isolated elements, such as the effect of dehydration on an aspect of power or endurance, may be limited by failing to capture some of the characteristics of weight category sports that can either mitigate or accentuate the performance effect. Others that include recovery strategies to overturn some of the negative outcomes of RWL may be more ecologically valid, but still fail to capture elements of performance or characteristics, such as repeated performance bouts, the effect of environmental conditions, or the need to weigh-in several times in a multiday competition. Nevertheless, an overall conclusion from this literature might be that RWL activities cause some direct decrements to elements of sports performance (Table 4), but these can be at least attenuated if there is sufficient time after the weigh-in to allow recovery nutrition strategies.

A major complication of defining the effect of RWL on performance is the concept of relative performance, which notes that competition success is awarded to the athlete who is best within their event pool rather than to the athlete who achieves their best performance level on the day of the event. The culture and experience of many athletes in weight category sports is that although RWL may reduce an athlete's absolute performance, RWL allows them to compete in a lower weight division where they may have a physical or psychological advantage, especially in events where RWL or the chance to recover after the weigh-in have minimal effects on performance. Yet, this is exceedingly challenging to quantify where it is most profound and relevant: in competition. Table 4 also includes studies of real-life attempts to observe correlations between the magnitude of RWL (measured or inferred from the BM increase after the weigh-in) and competition success. Even these studies have shown different outcomes, with findings spread across observations that athletes who appear to compete with the greatest magnitude of weight manipulation can be either less successful (39,125), equally successful (122,126,128) or more successful (36,129) than their counterparts who practice smaller degrees of apparent BM loss. Although further research is needed

Table 5.**Overview of strategies and rule changes to minimize harmful weight making behaviors in weight-category sports.**

Intervention	Positives	Negatives
Athlete and coach education	Required to help athletes make better dietary choices and improve awareness of harmful methods	May be ineffective without concurrent rule changes
Increasing the number of competitive weight categories	Competitors have more options to choose a weight class that is compatible with their habitual weight	- Creates a problem for sport organizations to ensure adequate number of competitors at each weight - Smaller weight increments between divisions create greater temptation to “cut” to a lower division
Use of height categories instead of weight categories	- Eliminates the need for RWL and RWG and the health risks associated with making weight - Maintains fairness in striking-based martial arts where limb-length is an important factor for competition success (<i>e.g.</i>, karate, taekwondo)	- Potential for large differences in body mass, strength, and power between competitors - Less suited to full-contact martial arts where use of body mass is important (<i>e.g.</i>, wrestling or judo) - Measurement error and diurnal variations in height could lead to misclassification
Establishing a minimal competition weight	- Encourages athletes to maintain their BM close to their competitive weight-category - Emphasizes chronic BM management through the manipulation of FFM and FM	- Estimation of BF% requires standardized equipment, methods, and trained personnel to ensure reliability - Requires a “season” or a standardized period each year where athletes are measured and certified - Athletes can still undergo large RWL prior to competition if they maintain a higher BF%
Move the weigh-in time closer to the start of competition	- Several sports with a short recovery duration (≤ 1 h) do not have issues with large RWL and RWG - Athletes undergo less RWL because of inadequate time to rehydrate/recover and the risk of a negative impact on performance in competition	- A risk that athletes may still undergo RWL and enter competition severely dehydrated because of inadequate recovery time—may increase health risk - Some argue that early weigh-ins are beneficial as it allows for maximum rehydration and mental preparation for competition
Determine the weight classes the day of competition	- Discourages any weight reduction because of unpredictability. - Effective administration at youth wrestling tournaments with abundant participants.	- Some weight discrepancies will exist within a weight class but no one is dehydrated or malnourished. - Difficult to administer for championship tournaments requiring prior qualifying tournaments.
Limit the number of weigh-in attempts during the official weigh-in period	Athletes who need to reweigh multiple times will likely use harmful practices (<i>e.g.</i>, spitting, vomiting, sauna, or training in rubber suits) in the short timeframe permitted before reweighing	To enforce the rule, athletes who miss weight at the first attempt must be excluded from competition
Assess urinary hydration status at the official weigh-in (specific gravity < 1.020 indicates adequate hydration)	- Discourages weight-making via dehydration-based methods by ensuring athletes are euhydrated at the time of the weigh-in - Easy to implement for sport organizations/events	- Concerns over the validity and reliability of field-based hydration tests: may lead to a high false-positive rate <i>i.e.</i>, hydrated athletes classified as dehydrated and being excluded from competition - Requires a calibrated refractometer at each event - To prevent fraud, each athlete must be closely observed when providing their urine sample (similar to anti-doping) - To enforce the rule, athletes showing dehydration must be excluded from competition
Competitors subject to a reweigh limit (tolerance) on the morning of competition	Discourages extreme RWL/RWG because of the need to remain within 5% or 10% of the weight category	- Still able to regain a large amount of body mass within the reweigh limit (<i>e.g.</i>, if 10%) - Competitors can maintain their weight below the limit and continue to rehydrate following the reweigh (typically 1 to 2 h precompetition, but can be longer) - Athletes may rehydrate, taking them over the reweigh limit, then perform another weight “cut” on the day of competition to be within the limit. - To enforce the rule, athletes who reweigh above the limit must be excluded from competition

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Intervention	Positives	Negatives
Prohibition of potentially harmful RWL methods (<i>e.g.</i> , IV rehydration, training in plastic suits, sauna use, laxatives, emetics, diuretics, etc.)	• Encourages athletes to use diet and exercise to slowly reduce their body mass for competition	• Challenging to enforce in the real world (athletes cannot be monitored 24/7) • In specific contexts, some methods deemed potentially harmful can be used in a safe manner

BF%, body fat percentage; IV, intravenous.

to gain better insights, it is recognized that outcomes are affected by methodological issues as well as real differences arising from the requirements of the sport. Regardless, there is evidence that at least some athletes sacrifice performance as well as incur health risks because of their weight making practices.

Programs and Rule Changes to Address Unsafe Weight Making Practices

In the 1960s, The National Federation of State High School Athletic Associations (NFHS) instructed U.S. states to introduce weight control plans to limit harmful RWL practices in high school wrestling (131). Initial attempts, focusing largely on athlete and coach education, had limited success (132,133), necessitating the development of structured programs alongside rule changes to make harmful weight-making impractical (see Table 5).

The first formal intervention was the Wisconsin Wrestling Minimum Weight Program (134), which led to a small, significant reduction in harmful weight-making practices in that state (135). Nevertheless, some wrestlers continued to undergo large RWL (>7 kg) for competition using undesirable methods (*e.g.*, training in heated rooms) and scores on a nutrition knowledge survey were no higher than those recorded before the program began (135). High school wrestlers subject to these rules also continued to engage in aggressive RWL during international-style wrestling tournaments (136), which occurred outside the scholastic season and its weight control regulations.

In 1997, following the tragic deaths of three wrestlers, the National Collegiate Athletic Association (NCAA) introduced a Wrestling Weight Certification Program (137), which expanded on the previous Wisconsin Program. The program: (i) limited weight loss to $\leq 1.5\%$ BM per week; (ii) determined a minimal competitive weight for each wrestler based on a lower limit of 5% body fat (males); (iii) moved weigh-ins to 1 h to 2 h precompetition; (iv) added 7 lbs (3.2 kg) to each weight category limit (*e.g.*, 190 lbs became 197 lbs); (v) prohibited the use of unsafe weight making methods; (vi) randomized the order of weight class competition (instead of heavier athletes competing later); and (vii) required athletes to pass a hydration test (urine specific gravity ≤ 1.020) at the weigh-in. When compared to practices observed prior to the NCAA Program (132,138,139), studies showed that wrestlers had less seasonal variation in BM (139–143), better retention of FFM (140), and a marked reduction in RWG (reduced to 1.2 ± 0.9 kg) between the weigh-in and competition (143). Taken collectively, the NCAA program reduced harmful weight-making practices to a greater extent than the Wisconsin program.

Despite this success, there are some concerns relating to the NCAA methods. The minimal competitive weight rule has been adopted by the U.S. Collegiate Sprint Football League (21), where male athletes must weigh under 178 lbs (81 kg) with no less than 5% body fat; and by the NFHS who set a higher minimum body fat (7% for males, 12% for females) to allow

Table 6.
Recommendations for safer weight making practices in weight category sports.

- Sporting organizations, practitioners/clinicians, coaches, and athletes involving events with weight categories should recognize the unique characteristics of their sport and support practices that promote safe and equitable competition for all competitors.
- Sporting organizations should continually review the weight making practices used within their sport to evolve the specific rules and regulations around weight categories and weigh-ins to deter unsafe practices.
- Continued evaluation of formal programs and sharing of best practices between weight-category sports are encouraged to fine-tune protocols based on success and failure to improve rapid weight loss practices, change the culture, and enhance safety and health of the competitors.
- Strategies should discourage any weight making among the minors and the youngest of participants and fully inform adult competitors of safety and health risks.
- The selection of a suitable weight category for each athlete, and the timely achievement of true alterations in BM towards the event target should underpin all weight making practices. A suitable weight category is one that can be safely achieved by the athlete without undue physical, nutritional or psychological stress.
- If formal weight-category selection programs have not been implemented, clinicians are encouraged to evaluate the athlete's body composition, not BM alone, guide weight class selection, and provide the caveat that minimal weight may not be the optimal performance weight.
- The athlete's natural or day to day BM should be within reach of their specific weight class, allowing them to achieve their usual training goals while maintaining dietary practices that support adequate energy availability, requirements for all nutrients, and a healthy relationship with food and its contribution to growth, development, and the quality of life.
- Critical planning and education are required to integrate nutrition needs with the training volume to ensure maintaining nutrient status while succeeding with FM reduction, maintaining FFM as much as possible, and improving competition readiness

for sex differences and maturation in adolescents (131). However, these standards have not been enforced across the United States and there is no guarantee of healthy weight management in high school or collegiate sports. Another concern is the stark differences in methods and procedures used to determine minimum competitive weight between states (144), which undermines the validity and reliability of the measurements. The use of urine specific gravity to measure hydration status at the official weigh-in also has limitations: current cut-offs can identify euhydrated athletes, but false-positives may be common, leading to athletes being incorrectly classified as dehydrated and excluded from competition (145,146).

Interventions like the Wisconsin and NCAA programs are challenging to implement for sports that lack the organizational structure of the US high-school and collegiate system. However, other combat sports have started to address the issue. Following a death in 2015, ONE Championship — Asia's largest MMA, Muay Thai, and kickboxing promotion — implemented a series of rule changes similar to the NCAA Program (147). More recently, the California State Athletic Commission introduced a 10-point plan in an attempt to reduce RWL to less than 10% (148). While these programs are promising, there are currently no data to show their impact on athlete weight-making behaviors.

A novel approach to eliminate weight making is the replacement of weight categories with height categories (1,37), which may be a viable approach for striking-based combat sports where limb-length is an important factor for competitive success. Other sport organizations have opted for simple changes to the weigh-in structure. Both World Taekwondo and the International Judo Federation now implement an evening-before competition weigh-in with a reweigh limit on the morning of competition (12,15). A sample of athletes complete the reweigh, with those weighing >5% above the upper limit of the weight category excluded from competition. The International Boxing Federation (IBF) applies a similar rule to championship bouts, with a reweigh limit 10 lbs (4.5 kg) above the upper limit of the weight category (10). A retrospective analysis of 77 IBF bouts showed the average RWG ranged from -0.3 to 6.4 kg (125), but it did not specify how many athletes exceeded the reweigh limit or if they were excluded from competition. While it is a common rule change, the effect of a morning reweigh limit on weight making behaviors needs further evaluation. A criticism of the rule is that athletes can easily manipulate their BM to stay within the reweigh limit, then continue to regain BM between the reweigh and competition (Table 5).

Despite several decades of work, harmful weight-making still exists in weight-category sports. However, resistance to new ideas should not be a deterrent to change: prior to the implementation of the Wisconsin program, most coaches (>60%) were opposed to it, but four years later nearly all (>95%) coaches supported it (134). New programs and regulations should involve all key stakeholders, be pilot tested, evaluated, and revised, as necessary. These then need to be implemented by the largest sport organizations and adopted downstream by national and regional federations to have the greatest impact.

Summary

A variety of sports involve weight categories with the aim of organizing competition that is safe and equitable between

competitors with similar physical characteristics. Athletes in such sports use chronic and acute weight making techniques to compete in a weight category that is lower than their “natural” weight, gaining a theoretical advantage over smaller opponents. Extreme weight making practices, however, often lead to health and performance concerns. The ingrained culture and practices of weight making are difficult to remove and should be recognized as being unique to the specific characteristics of each sport. While no single approach is likely to be successful across all weight category sports, guidelines underpinning programs to address concerns around weight making can be provided in principle (see Table 6). Strategies to support education around safer practices include minimal competition weight classification based on preseason body composition, reductions in the period between weigh-in and competition, and prohibition of unhealthy weight loss techniques. At an individual level, expert guidance by sports nutrition professionals can help an athlete to establish a pragmatic and long-term approach to BM management, recognizing the nuances of their sport, to achieve favorable outcomes for both health and performance.

Click here (<http://links.lww.com/CSMR/A111>) to download a slide deck that summarizes this ACSM Expert Consensus Statement on Weight Loss in Weight-Category Sports.

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